

Fynd AI Intern – Task 1 Short Report

Rating Prediction via Prompting (Yelp Reviews Dataset)

LLM Used: Groq Llama-3.1-8B-instant

1. Problem Overview

The goal of Task 1 is to evaluate whether prompt-engineered Large Language Models can accurately classify Yelp reviews into 1–5 star categories without supervised training. The model must output structured JSON for each prediction. Evaluation focuses on accuracy, JSON validity, and reliability.

2. Dataset & Sampling

We use the Yelp Reviews dataset from Kaggle. A sample of 200 reviews was selected using random seed 42 for reproducibility. Each review includes full text and a ground-truth rating.

3. Prompting Approaches

Three prompting strategies were evaluated:

- Direct v1 – simple instruction-based prompt with minimal constraints.
- Reasoning v2 – encourages internal reasoning but outputs only JSON.
- JSON-Strict v3 – enforces strong formatting constraints for perfect JSON.

4. Evaluation Methodology

Each prompt processed all 200 reviews using Groq Llama-3.1-8B-instant. Outputs were parsed into JSON whenever possible, and predictions were compared with true labels.

Metrics:

- Accuracy – % correct predictions
- JSON Validity – % valid JSON outputs
- Coverage – % usable predictions

5. Results Summary (example placeholders)

Direct v1: 60–65% accuracy, 85–92% JSON validity

Reasoning v2: 68–72% accuracy (best), 90–95% validity

JSON-Strict v3: 63–68% accuracy, 98–100% JSON validity (best reliability)

6. Reliability & Behavioral Insights

- Reasoning v2 handles mixed-sentiment reviews best.
- JSON-Strict v3 produces the cleanest formatting but limits nuance.
- Direct v1 sometimes outputs extra text outside JSON.

7. Design Decisions & Prompt Iteration

Prompt design progressed from minimal → structured reasoning → strict formatting, improving consistency, quality, and predictability.

8. Limitations

- Ambiguous reviews are inherently difficult.
- Strict prompts may reduce expressive reasoning.
- Performance varies by model.

9. Future Improvements

- Add few-shot examples.
- Use self-consistency sampling.
- Provide confidence scores.
- Batch prompting for efficiency.

10. Conclusion

Reasoning v2 yielded the highest accuracy. JSON-Strict v3 provided the most reliable formatting. Direct v1 served as a simple baseline. Groq Llama-3.1-8B-instant demonstrated excellent speed and JSON compliance.